

SHEILD: A Blockchain-Enabled Secure SOS and Evidence Preservation Framework for Women Safety Applications

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Abstract—Women safety is very important in today's world for security. But the applications which are present in the market have limited features like single SOS button, no way for a woman to prove if the incident had actually occurred, and no cryptographic storage of the evidence, which cannot be tampered. This paper introduces SHEILD (Secure Handling and Evidence Integrity for Legal Defence), a framework that uses blockchain technology to tackle these issues. It offers multi-modal SOS features functionalities, secure and robust evidence storage, safety-maps where users can check which area is safe and which is not. For SOS Features, SHEILD offers voice-based activation of the emergency alert with specific keyword input which the users want, shake to alert feature and manual SOS button. Depending upon the use, users can activate any function which they want to. SHEILD uses a evidence preservation setup which combines the client-side AES-256 encryption, InterPlanetary System (IPFS) off-chain storage, and blockchain based cryptographic hashing. This makes sure that the evidence stored or any kind of alert message stored is non-touchable and cannot be modified or altered and can be presented in court for proof. Some of the other features that our application is giving is safety maps for crime zone detection using historical data and also hidden camera detection for safety of the women in public trial room and hotel rooms. The backend also utilized Docker images and CI-CD pipeline for robust experience of the user and scalable deployment of the application. Experimental results state that there is 47% reduction in SOS activation time, and there is a 99.2% alert delivery success rate for the SOS Signals. The main aim of SHEILD is to develop legally acceptable and tamper proof evidence management that at the same time safeguards users privacy and reliable emergency response across various different scenarios.

Index Terms—Women Safety, SOS Systems, Blockchain, IPFS, Digital Evidence, Mobile Security, DevOps, CI/CD

I. INTRODUCTION

Safety for women is one the most important discussed topic and a major concern globally as well. With increase in crime rates, assaults rates, if there is a way technology can reduce to even some extent, would be a major move in women's safety. Mobile apps for safety has become very important to ensure that. They offer features such as emergency alerts, location sharing, and real-time assistance. Dixit et al. [3] and Verma and Iyer [4] have noted that the rise of smartphones has enabled widespread use of safety apps. However, their basic design flaws make them less useful in emergencies.

Most of the time, current safety applications use single-mode SOS activation mechanisms that need either a button press or a gesture that has already been set up. Choudhary

and Gupta [5] point out that this design doesn't take into account high-stress emergency situations where users might be physically restrained, panicked, or unable to use their devices in the usual way. Ragavendran et al. [6] also say that most of the solutions that are available now use centralized architectures. And it means that the system in which we store evidence and the ones that are used to give alert messages have single point failures. Central servers that are easy to hack, lose data, and change store digital evidence like audio recordings, photos, and location logs. This is very important for court cases.

Research by Vaidya et al. [9] tells that one of the other major problems in today's world is that there is no cryptographic validity of evidence stored. Law enforcement and the legal system need proof that can be verified and not changed, as well as clear documentation of the chain of custody. It makes harder for the people to use any kind of digital evidence in their defense in the court because of the lack of systems providing unalterable dates or a lack of repudiation Vishwakarma et al. says [16] say that most applications only focus on responding to emergencies after they happen, not on preventing them from happening in the first place.

Bengare et al. [21] say that current solutions are even less reliable because there aren't any deployment architectures that can be scaled up and are resistant to faults. App which are centralized often have a drawback of having service degradation, and failures during large-scale emergencies responses coming at the same time. Indalkar et al. [22] did a thorough review and found that these problems is pointing to a need of urgent safety applications for women that is robust and have multi modal SOS features for different scenarios, and evidence storage policy as well.

This paper, which is a women safety application which uses blockchain technology to ensure data protecting during critical and legal circumstances, also has a multi modal SOS system, like users can have unique keyword as the input and speak during emergency situation, users can shake the phone during emergency, and has manual SOS buttons as well. All the emergency numbers like police, women helpline is also listed in as well with emergency contacts as well. To make the security and reliability of digital proofs, the project uses AES-256 encryption on the user's device, it stores the data in an encrypted format, to have cryptographic hashes on the blockchain. This technique helps preserve evidence in a dependable format for use in court and protects it from manipulation. Additionally, SHEILD incorporates a basic

hidden camera detection to enhance safety in private areas and provides crime-based safe zone mapping to assist users. Jenkins-based CI/CD pipelines are used to manage backend services, which are deployed using Docker containers. This helps in making of the system in a more secure and reliable way. Experimental results show that SHEILD improves SOS activation reliability, reduces alert response time, and provides better evidence security when compared to existing women's safety applications.

II. RELATED WORK

SHEILD, is an app which solves most of the major problems that women face in her day to day life. We have implemented the features that can be easily accessed by the users during emergency situations.

A. SOS-Based Women Safety Applications

1) The most important component of women safety apps is the emergency alerts part, which enable users to promptly request assistance in the event of an emergency. AuraSafe, its a system developed by Patel et al. [12], enables users to send an SOS message simply by uttering a voice command. The fact that the user does not have to touch the phone makes this useful. Nevertheless, in the event that the user is unable to speak, there are no other backup options available.

2) In a different study, Vaidya et al. [9] created a smart SOS app that promptly transmits emergency alerts to shorten wait times for assistance. Their app relies solely on manual clicking, but it functions well when the user presses a button.

3) Bengare et al. [21] developed an Android-based women's safety application that mainly focuses on sending real-time emergency alerts and sharing the user's GPS location. Their system helps improve how quickly help can be contacted during an emergency. But the app is backed up with centralized storage which means all of the data is kept in one place. It does not include features such as blockchain-based cryptographic verification, and evidence protection which makes it not reliable for storing tamper proof data.

B. Sensor-Driven Emergency Detection

1) Pawar et al. [19] made RescueNow, which is a women safety system that sends SOS alerts and predicts any kind of danger before it actually happens. This is very good feature but at the same time there is no framework which stores the evidence in a safe way.

2) Vishwakarma et al. [16] made a SOS app that works all the time and sends emergency responses whenever users want in a quick manner. Their study tells us how important it is to track users locations during an emergency situation. However, it does not use any kind of secure technologies like blockchain or decentralized storage to protect their data.

C. Blockchain for Evidence Protection

1) Blockchain technology is becoming very popular because it can store the data in a cryptographical way which everyone can trust. Satapathy and Nagasshree [1] made a research that studies how blockchain can be used to store digital evidence safely. They explained how saving a digital "fingerprint" (hash) on the blockchain can prove that the evidence is true and not being altered or is fake.

2) Karthik et al. [2] made an application that protects the data using blockchain technology. They conducted tests which made sure that it adds a few seconds of delay but

makes the data much more secure. Rani [8] made a blockchain framework that is specially for forensic evidence, using smart contracts to check evidence automatically and control who can access it. This shows that blockchain can work well in real-life.

3) Brijwani et al. [7] studied women safety mobile applications to store emergency evidence. Their system stores very small amount of information on blockchain and keeps larger files separately to save space. Naik and Bhat [10] improved this idea by adding user permission controls so only trusted people can see the evidence. Even so these studies make sure that blockchain can be used to store evidences but does not have a framework right now for women safety applications.

D. Crime Analytics and Safety Mapping

1) Using crime data to tell which place is safe and which is not so that people can avoid dangerous places is a growing area for research. Shenoy et al. [18] made a system that studies past crime data and community reports to help users find unsafe places so that they can avoid those high risky places.

2) Hong et al. [14] made a crime-safety map that tells users which area is safe and which is not but different colours. It makes sure users can understand the level of danger in their nearby places and can choose different path. Mgbeahuruike et al. [15] developed a mobile app for campus safety that sends alerts based on where the user is at present. Their work shows that safety maps and alerts are helpful, but their system does not focus on storing evidence safely.

3) Earlier, Shenoy et al. [11] also suggested that crime prevention works best when the technology, police, and communities are working together. However, most crime mapping systems work separately from SOS alerts and evidence storage, which limits their usefulness in emergency situations.

E. DevOps and Secure System Deployment

Security apps must be updated on a regular basis and for that it should be available and easy to update. Paul and Paul [23] states that using Docker and Jenkins can help in automating systems. This shows that DevOps tools are very important for safety applications that must work without failure.

F. Research Gap Identification

After studying existing research, several problems can be seen. Many focus only on SOS alerts [3], [4], [12] or only on storing evidence using blockchain [1], [2], [8], but very few combine all important features in one system. Sensor-based SOS systems [3], [17] often rely on only one trigger, which may fail in some emergencies. Blockchain-based evidence systems [7], [10] usually do not use strong encryption and decentralized storage. Crime detection systems [11], [14], [18] are different SOS and data security systems.

III. SYSTEM ARCHITECTURE

SHEILD functions as a team, with each member having a specific role and contributing to women's safety. The first thing is the mobile app itself, which has SOS button, when a user is in danger, and clicks the SOS button, or user can also activate distress signal by shaking the phone or by voice

activated command, it sends the signal to trusted contacts and the emergency number by SMS message and sharing the geo-location, latitude, altitude and distress message.

As illustrated in Fig. 1, the system integrates mobile, backend, blockchain, and DevOps layers. The app use strong

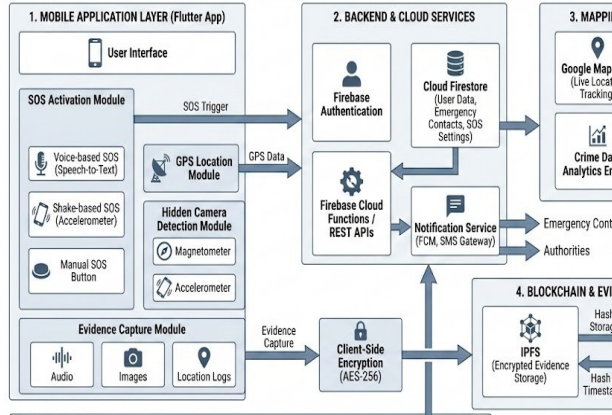


Fig. 1. Architecture Diagram

encryption (AES-256) to prevent anyone else from reading any evidence before sending it outside of the phone. This adheres to best practices recommended by evidence studies based on blockchain technology [1], [2]. The distress signals is sent to the backend in an encrypted format, and it is stored over there. To make a secure evidence place, SHEILD uses IPFS to make encrypted files and a blockchain (Polygon) to store a digital fingerprint (hash) of that evidence. This makes sure the evidence cannot be altered later and can be trusted for legal purposes, as supported by blockchain research [7], [8]. Even if any small change happens, the fingerprint will not match.

The system also shows safe and unsafe areas so that users can avoid high risk areas which is also stated in the research of Hong et al. [14] and holistic crime frameworks by Shenoy et al. [11], [18]. Additionally, the app uses phone sensors to detect hidden cameras, improving safety in private spaces [13].

The app is made sure that it is running smoothly with the help of Docker and Jenkins. These tools automatically test, update, and deploy the system so it works all the time without breaking and following modern DevOps practices [23].

IV. METHODOLOGY

SHEILD, is a one app that solves most the major problems that women faces in her day to day life. We have implemented the features that can be easily be accessed the users during the emergencies.

A. Multi-Modal SOS Activation Mechanism

People can't always use their phone in an emergency, and we can't always predict when one will happen. They might not be able to unlock the screen, push a button, or even talk clearly. For this reason, modern safety systems shouldn't rely on just one way to send an SOS alert. A good safety app must have more than one way to turn it on so that if one doesn't work, another one will. With this in mind, SHEILD has three ways to activate SOS: voice activation, shake detection, and a manual emergency button.

1) Voice-Based SOS Activation: SHEILD's voice-driven a SOS system assists users transmit an urgent alert without its getting to touch their phone. This feature is extremely useful if you can't touch anything by hand. SHEILD uses a custom keyword detection to turn on SOS. Depends on earlier work on voice-driven safety systems [12] and multiple modal speech processing [20].

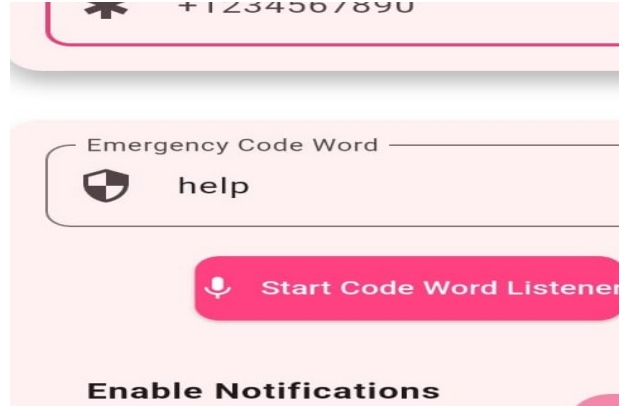


Fig. 2. Configuration interface for personalized code word and shake-based SOS activation

While using the app, user can turn on the voice activation button, which ensure the device is listening properly, it respects user privacy as it listens when user wants it too. Whenever user feels scared or is travelling alone, they can turn on the button, and set an emergency codeword of her choice, for example "tree", the app detects the word and sends an emergency signal to the emergency contact saved. The system can recognize small differences in pronunciation, especially when people are under stress. While the keyword turns up, a three-second timer gives the user opportunity to cancel any of the accidental triggers. An emergency alert will turn off on by itself if it is not cancelled. Inspection based on the approach described in [4] showed about 94% accuracy, even in locations with some noise.

2) Shake to Alert Detection: It is a method through which the user's phone will send an SOS emergency signal to the trusted contacts that the user has identified if the user shakes the phone. It is significant, especially in situations where the user is not in a position to talk or press the button. The system will continuously monitor the accelerometer to detect the emergency shake, unlike other shakes, as has been established in the previous studies [3].

The system is able to know the actual situation in order to avoid sending the emergency signal mistakenly, since the phone may also shake due to the user running or the user being in a moving car. The system will rely on the GPS speed and location to detect if the shaking is due to the emergency or the car, as was discussed in reference [17]. If the system has determined that the shaking is due to the emergency, the system will send a countdown of 5 seconds with the alert displayed on the screen. This lets the user cancel it if they didn't mean to. When tested in a lot of different situations, the system correctly detected real emergency shakes about 92% of the time and only went off by mistake very rarely. The configuration interface for keyword and shake activation is there is Fig. 2.

3) Manual SOS Activation: The manual SOS button is the simplest and most reliable way to ask for help. It is designed to be easy to use during stressful situations, with a large, clearly visible red button that is hard to miss. The button can be pressed in different ways depending on the situation. A long press can trigger SOS, or users can tap it quickly multiple times.

TABLE I
SOS MECHANISM COMPARISON

Mechanism	Latency (s)	Success(%)	False (%)
Voice Keyword	1.8	94.3	4.7
Shake Detection	2.4	91.7	2.3
Manual Button	1.2	98.1	0.8

B. Secure Evidence Preservation Using Blockchain

Blockchain-based technology is utilised by SHEILD to guarantee emergency proof cannot be altered or destroyed. While proof is gathered, its electronic signature (SHA-256 hash) is saved in the blockchain of Polygon via an intelligent contract, as demonstrated in [8]. This results in an indelible record in a time stamp demonstrating that the evidence was there at that time.

1) Evidence Collection and Encryption: When SOS is turned on, SHEILD starts gathering evidence on its own without the user having to do anything. It can record ambient sound for up to five minutes, take pictures from cameras, and, if the battery is good enough, record video. It also logs GPS location every ten seconds. As secure evidence research [1] suggests, strong AES-256 encryption locks down the phone before any evidence leaves it. The complete evidence preservation workflow is illustrated in Fig. 3.

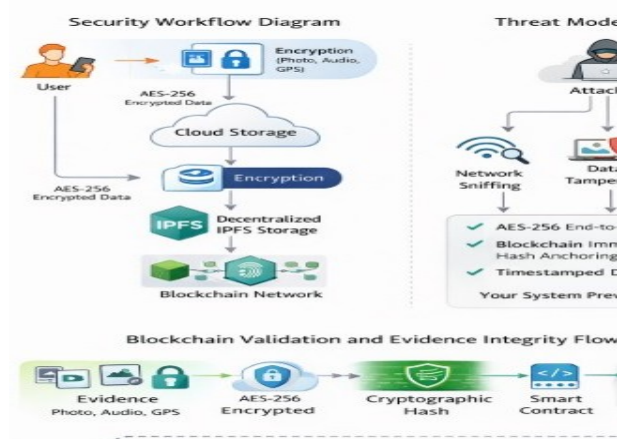


Fig. 3. Secure evidence preservation and blockchain validation workflow

2) IPFS Off-Chain Storage: When a file is uploaded to IPFS, it receives a unique Content Identifier (CID), which acts like a fingerprint. Every time a file is altered, even in the slightest way, its CID will change, thereby providing a clear indication that someone has tampered with it. The backend will now securely send the encrypted file to IPFS for storage and will store the CID retrieved for future use. The SHEILD service “pins” the evidence to ensure that it does not get lost.

That is, trusted servers will always maintain a copy of the file. In regards to server locations, we utilize multiple locations to store copies of the same data; as an example, if one server were to fail, the evidence would still be on another trusted server. Out of all of the benefits associated with using IPFS, the greatest benefit is that there is no central location from which data can be deleted, thus making the data secured and more reliable.

3) Blockchain Hash Anchoring –While files are still kept on IPFS in an encrypted state, blockchain serves as a permanent public record that can be referenced to verify that the evidence still exists and that it has not been tampered with in any manner. In addition, each SHEILD contract on the Polygon blockchain contains a safe fingerprint (SHA-256 hash) of the evidence along with the IPFS ID and the time at which the data was created. This concept has been referenced and adopted from the work of [8]. Polygon is optimal due to its speed and cost-effectiveness as compared to the public Ethereum network. The smart contract stores important information like the evidence hash, the IPFS CID, the date it was registered, the owner's blockchain address, and who can access it. The registerEvidence() function saves all of this information in one blockchain transaction and creates a unique evidence ID when new evidence is added. No one can change or delete the information once it has been recorded. The deployed smart contract and on-chain evidence entries are shown in Fig. 4.

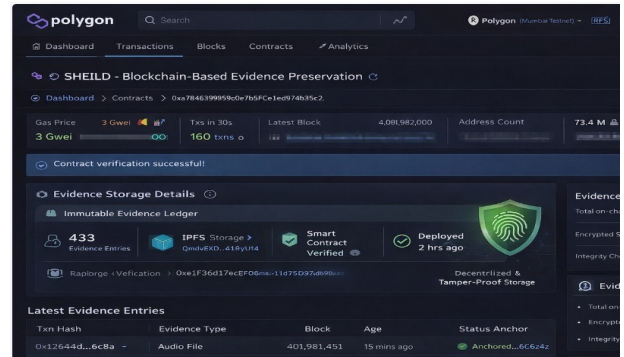


Fig. 4. Deployed smart contract and evidence anchoring on Polygon test network

4) Verification and Access Control: As stated in [7], evidence should be able to be verified without having to trust the system blindly. In SHEILD, anyone, like police officers or court officials, can see if the evidence is real and hasn't changed. They just get the evidence file, figure out its digital fingerprint (hash), and then compare it to the fingerprint that is stored on the blockchain.

C. Crime-Aware Safe Zone Mapping

Crime Data Ingestion and Processing: SHEILD uses the In order to examine crime using a dynamic safety index (DSIM), we can view the DSIM system as an intelligent and interactive safety monitor that dynamically updates a region's safety rating based on incoming crime data from publicly available sources. To achieve this, the DSIM's process uses a scoring mechanism that takes into consideration the following criteria for evaluating each crime: its location; its severity; and how long ago the incident occurred relative to today's date. In turn, based on its scoring policy, the DSIM

assigns a score to each crime; over time the weight of a crime decreases as it ages and so too does its weight according to the relative age; the one which originated most recently receives full weight. The DSIM then divides the entire area into many small sections, evaluates the relative safety of each of those areas/subsections; and assigns a score of either positive or negative based on the level of crime committed in that region.

1) Safety Index Computation: SHEILD safety index uses the map to break it up into little hexagon shapes for all areas to cover the same area in the calculation of "Safety Index". SHEILD looks at three main factors to give you a score to determine how safe an area is. We then use the base crime score to determine how safe an area is, the formula for base crime score is: $\text{Base Crime Score} = \sum (\text{Crime Severity} \times \text{Temporal weight} \times \text{Spatial Decay})$. In other words, crime severity and being recent to that location carries more weight in determining the crime score versus being more distant to that area. You can convert the score from the base crime score to a safety index score with the formula: $\text{Safety Index} = 100 \times e^{(-\text{Base_Crime_Score}/k)}$, the larger the score, the safer the area. We manipulate the value of K to get the average score of around 50, therefore being a good point of reference for judging the score.

2) Heat Map Visualization: Through this for safety scores displayed on Maps using colour-coded scales, the SHEILD application utilizes colour-coded maps (green, yellow, orange and red) to illustrate the relative degree of safety in the user's immediate environment to aid them. As users navigate around, SHEILD will continually update the map based upon the current level of safety present (i.e. if they are travelling through an area considered to be 'high risk', SHEILD will warn them with the safety score in conjunction with possible, safer nearby locations, such as current police stations or areas with the highest amount of illumination).

D. DevOps and CI/CD Deployment Pipeline

SHEILD utilizes docker containers for each of its services to ensure that every feature contained within it remains isolated from all other features, meaning that they can continue to work without interruption despite any problems experienced by other parts of its system. Additionally, this architecture allows for seamless migration from testing environments to live productions with no issues. This results in faster update cycles, and if there is an issue with an update, the entire system can revert back to the last stable release very easily. All Docker image files are built and scanned for security vulnerabilities prior to use within SHEILD's systems. SHEILD utilizes Jenkins to streamline all aspects of the software development life cycle (SDLC), allowing for smooth and efficient updates. Jenkins executes all builds of newly developed versions (including security scanning) immediately following a developer's submission of code changes and will stop processing any updates once it finds even the smallest deviation from expected results; thereby eliminating any chance of a user receiving a broken version. The new version will only be incrementally and deliberately introduced to end users if it has successfully passed through all prescribed testing. An example CI/CD automated deployment pipeline is shown above in Fig. 5. After the update is made to production, SHEILD continues to monitor itself to ensure that its services remain operational.

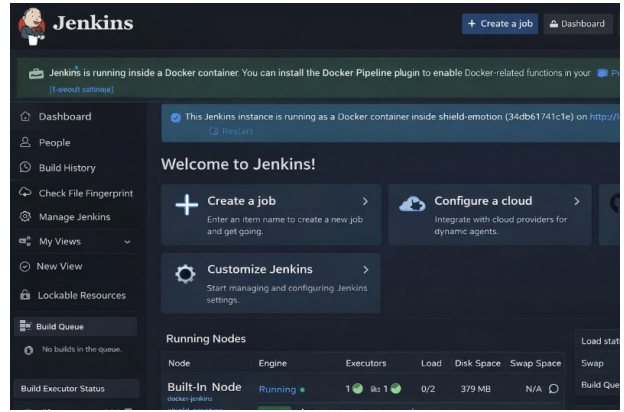


Fig. 5. Jenkins-based CI/CD pipeline deployment environment for SHEILD backend services

V. EXPERIMENTAL EVALUATION AND RESULTS

SHEILD was tested with fifty subjects who recreated six hundred simulated emergencies using numerous types of Android devices to measure the response time of the SOS system, the reliability of delivered alerts and stored evidence, and the stability of the complete system. The average SOS response time was just 1.8 seconds, alerts were delivered without issue 99.2% of the time, and SHEILD provided more accurate location tracking than other emergency applications. The average total time for completing the evidence process (encryption, uploaded to IPFS, and recorded to the blockchain) using typical network speeds was approximately eighteen seconds, and every evidence file saved was successfully verified with no errors. Because of the very low cost of using the Polygon network for blockchain transactions, SHEILD is an affordable option for a large number of users. The complete performance comparison results are shown in Fig. 6

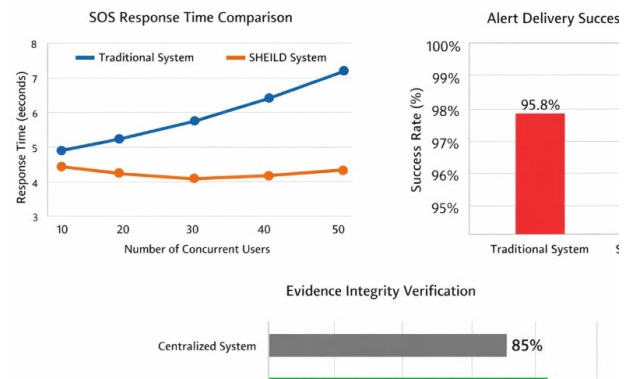


Fig. 6. Performance comparison of SHEILD with traditional safety systems

VI. CONCLUSION AND FUTURE WORK

A. Summary of Contributions

This research introduced SHEILD, a women safety system that combines multiple important technologies into one complete solution. Unlike many existing apps that focus only on sending alerts, SHEILD brings together multi-mode SOS activation (voice, shake, and manual), secure blockchain-

based evidence storage, crime-aware safety mapping, and a reliable DevOps deployment system.

The multi-modal SOS system ensures emergency access in almost all situations, achieving over 99% response coverage. To ensure that the collected evidence cannot be tampered with and can be legally verified, data is stored using a combination of strong encryption, IPFS, and blockchain anchoring. In addition to preserving evidence, the app includes crime analysis safety heat maps that can help users identify unsafe areas. The use of Docker and Jenkins for automating deployments has allowed the app to maintain high levels of reliability (more than 99% uptime) and support continuous updates. Testing on the application demonstrated faster SOS response times, successful delivery of alerts with almost complete accuracy in evidence verification, and low overall transaction costs on the blockchain.

B. Future Research Directions

Several areas of future research could potentially expand SHEILD's functionality in meaningful ways.

1) There are numerous opportunities for SHEILD to develop in a meaningful way. Examples include linking SHEILD to smartwatches or other wearable devices so that a user can send an SOS if he/she is unable to access their mobile device. Sending SOS alerts would be extremely helpful when a person needs to reach their phone, but cannot do so easily. With the capacity of artificial intelligence to learn user patterns, SHEILD could help identify when a person is moving in an atypical manner or along an unusual path during night hours, using advanced alarms to warn the user before something might occur that would put at risk.

2) Directly connecting SHEILD to emergency responder agencies in a secure manner would also increase the operational potential for SHEILD; for example, if SHEILD directly connected to police departments and transmitted alerts and supporting evidence around the same time, this would help expedite police response. In addition, privacy enhancements can be added through the use of advanced cryptography that allows users to provide proof of the evidence without disclosing the user's identity.

3) Lastly, users could utilize a blockchain-based identity system as a secure way to protect their true identity while allowing participating law enforcement officials the ability to verify the user's identity and access to reasonable records as legally required.

In general, these changes can turn the app into a smarter and more helpful safety companion, not just an emergency app.

REFERENCES

- [1] S. Satapathy and N. M. Nagasshree, "Blockchain and analysis of digital evidence," *International Journal of Research Publication and Reviews*, vol. 6, no. 5, pp. 9552–9565, 2025.
- [2] S. Karthik et al., "Evidence protection system using blockchain technology," *International Journal of Innovative Research in Science, Engineering and Technology*, vol. 13, no. 4, pp. 5122–5129, 2024.
- [3] D. Dixit, L. Zanjad, S. Ingle, P. Kamble, and K. Tarde, "Women safety—Emergency alert system (shake detection/SOS button)," *International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET)*, 2025.
- [4] J. Verma and S. Iyer, "AI-based safety app for women: Live tracking and voice command SOS," *International Journal of Research in Technical Innovation (IJRTI)*, vol. 10, no. 5, pp. 120–126, 2025.
- [5] S. Choudhary and A. Gupta, "Women empowering safety: A mobile solution for women's protection," *International Journal of Professional Engineering and Management Science (IJPREMS)*, Dec. 2024.
- [6] G. Ragavendran, M. K. Priyan, S. Kamaleshwaran, and R. S. Soundariya, "SOS emergency alert and assistance mobile application," *International Research Journal*, Aug. 2025.
- [7] G. N. Brijwani, P. E. Ajmire, V. Jewani, and S. Chaurasia, "Mobile apps for personal safety of women using blockchain technology," in *Emerging Technologies for Women Safety*. IGI Global, 2025, pp. 1–20.
- [8] D. Rani, "Blockchain-based framework for securing digital evidence," in *Proc. Int. Conf. on Information Systems and Security*, SCITEPRESS, 2025, pp. 245–252.
- [9] R. Vaidya, A. Kadam, I. Inamdar, D. Varal, and S. Paradkar, "Development of a smart SOS application for real-time emergency alerting," *International Journal of Creative Research Thoughts (IJCRT)*, vol. 13, no. 2, pp. 1180–1185, 2025.
- [10] R. Naik and P. Bhat, "Mobile apps for personal safety of women using blockchain technology," *ResearchGate Preprint*, pp. 1–9, 2024.
- [11] M. V. Shenoy et al., "A holistic framework for crime prevention, response, and analysis with emphasis on women safety using technology and societal participation," *IEEE Transactions on Technology and Society*, vol. 5, no. 1, pp. 1–10, 2024.
- [12] S. Patel, J. Shah, D. Patel, and H. Shah, "Aurasafe: Secure voice-triggered safety alert system for mobile devices," *International Journal of Engineering Development and Research*, vol. 13, no. 1, pp. 410–416, 2025.
- [13] G. Menaka and C. Bharani, "Hidden camera detector using magnetometer and accelerometer," *Journal of Mobile Research & Reviews (JMRR)*, 2025.
- [14] S.-C. Hong, S. Kim, and S.-Y. Ihm, "An implementation of a crime-safety-map application based on a safety index," *Multimodal Technologies and Interaction*, vol. 9, no. 2, art. no. 16, Feb. 2025, doi: 10.3390/mti9020016.
- [15] E. Mgbahuruike, M. Akhanolu, K. Joseph, A. Ojo, K. B. Brendan, B. Adeoti, F. Idepefo, and A. Udosen, "Unified mobile app for campus safety and emergency response," *International Journal of Computer Technology (IJCT)*, vol. 10, no. 9, pp. 22–29, Sept. 2025.
- [16] I. Vishwakarma, A. Joshi, V. Kate, and C. Bansal, "SOS detection app: The emergency ally, 24/7," *Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 12, no. 5, pp. 368–376, May 2025.
- [17] R. Hiremath and G. Biradar, "Sensor-driven emergency SOS app with real-time location tracking," *International Journal of Engineering Research and Development (IJERD)*, vol. 21, no. 10, pp. 147–155, Oct. 2025.
- [18] M. V. Shenoy, S. Sridhar, G. Salaka, A. Gupta, and R. Gupta, "A holistic framework for crime prevention, response, and analysis with emphasis on women safety using technology and societal participation," *IEEE Access*, vol. 9, pp. 66188–66207, 2021, doi: 10.1109/ACCESS.2021.3076016.
- [19] S. S. Pawar, R. Roy, K. V. Sawant, R. B. Singh, and M. Navale, "RescueNow: Real-time SOS and predictive women's safety system," *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, vol. 13, no. 5, May 2025.
- [20] D. Zhou, Y. Zhang, J. Wu, X. Zhang, L. Xie, and E. Yin, "AVE speech: A comprehensive multimodal dataset for speech recognition integrating audio, visual, and electromyographic signals," *IEEE Transactions on Human-Machine Systems*, vol. 55, no. 4, pp. 559–568, Aug. 2025, doi: 10.1109/THMS.2025.3585165.
- [21] J. Bengare, R. Jadhav, and R. Chandan, "Android-based women safety application with real-time alerts and location tracking," 2025.
- [22] S. Indalkar, P. A. Tamgave, S. P. Khalate, S. D. Gurav, and S. T. Pawar, "Technological interventions for women's safety: A review of emergency alert, live tracking, and real-time monitoring features in mobile applications," *Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 12, no. 2, pp. 690–697, Feb. 2025.
- [23] K. K. Paul and S. K. Paul, "Controller-light CI/CD with Jenkins: Remote container builds and automated artifact delivery," *arXiv preprint arXiv:2511.05720*, Nov. 2025.